Quant Class Assignment 10

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April 2026

1 Introduction

This project analyzes the impact of small class sizes on high school graduation rates using the STAR dataset.

2 Methodology

To test this hypothesis, I estimated two Linear Probability Models. The first model is a simple bivariate regression where high school graduation is regressed on the small class assignment indicator. The second model adds complexity by controlling for student race and the total number of years the student spent in a small class environment.

3 Results

The regression results presented in Table 1 show the estimated effects of class size assignments on the probability of graduating high school.

Table 1: LPM Models of High School Graduation

[caption=Effect of Small Class Size on Graduation, note=+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001,] colspec=Q[]Q[]Q[], hline2=1-3solid, black, 0.05em, hline16=1-3solid, black, 0.05em, hline1=1-3solid, black, 0.08em, hline20=1-3solid, black, 0.08em, column2-3=halign=c, column1=halign=l,

	LPM biv.	LPM ctrl.
(Intercept)	0.832***	0.860***
	(0.008)	(0.009)
small	0.004	−0.076**
	(0.015)	(0.026)
raceBlack		−0.127***
		(0.015)
raceAsian		−0.024
		(0.150)
raceNative American		0.140
		(0.368)
raceOther		0.165
		(0.260)
yearssmall		0.027***
		(0.007)
Num.Obs.	3047	3047
R2	0.000	0.028
R2 Adj.	0.000	0.026
Std.Errors	HC3	HC3

As shown in the table, the "Small" class variable provides an estimate of the treatment effect. To better visualize these results, Figure 1 provides a coefficient plot showing the point estimates and their associated 95% confidence intervals

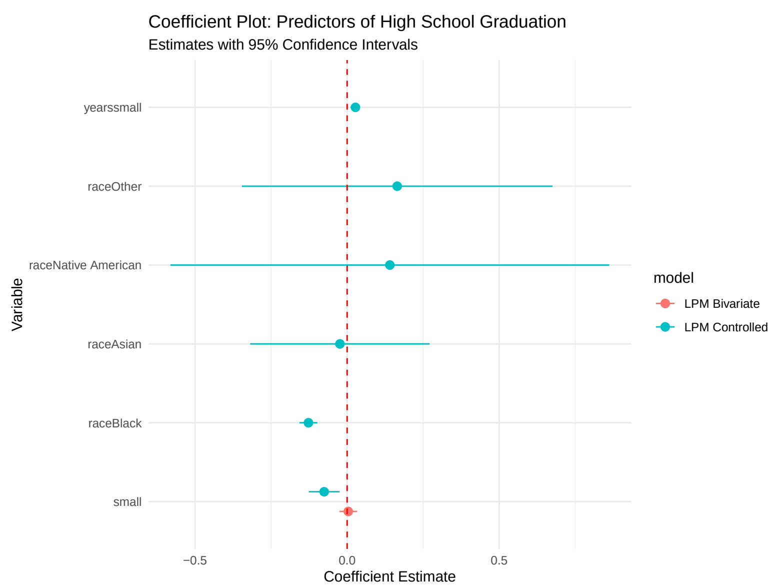

Figure 1: Coefficient Plot of Predictors for High School Graduation

4 Conclusion

This project demonstrates a reproducible research pipeline. When the data or model specifications change in the R script, this document can be updated automatically without the need for manual copy-pasting of results.